Open Source Coding - OPSC7311
Portfolio of Evidence
Part 2 - Design

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1. Introduction

With South Africa's cost of living crisis becoming more and more prevalent, it has become almost imperative to have a form of budgeting being done in any person or family in their day to day life. From dinner choices being limited to petrol for the cars being divided depending on usage per week/month, budgeting apps have become a new addition for more accountable budgeting as they keep better track of one's own finances which urges users to budget better. Apps like Wallet and Vault22 are predominantly free applications that have innovative features such as synchronisation to bank accounts and others, but suffer from a strict budgeting system and customer service/frequent bug issues. This is a design document for our own budgeting app based on our own understanding from previous research of budgeting apps.

2. Overview

Our budgeting app is a free/predominantly free to use application that offers a gamified financial management system while maintaining a sort of strict budgeting system by ensuring that transactions are not automated and users will have to log their expenses. Users can create short, medium, and long term (weekly, monthly, annually) budget plans and can keep track of said budget's progress with a percentage of budget spent, the amount left over to spend, and a recommended amount that should be spent daily to maintain the budget. Users will be able to set up multiple categories for their budgets as well as view their expenses to properly gauge their spending habits. There are also prizes to win when a user meets a budget goal (Eg. Going a month without exceeding a daily spending limit for a week).

Icon Design



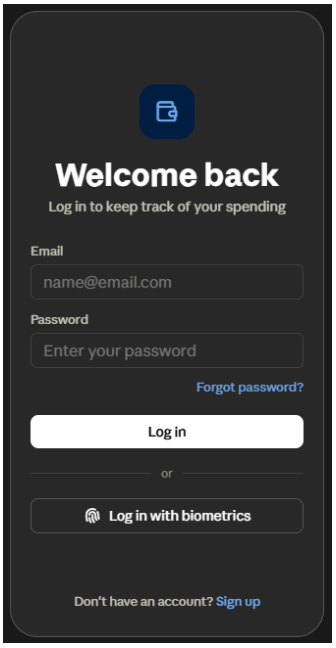
Innovative Features

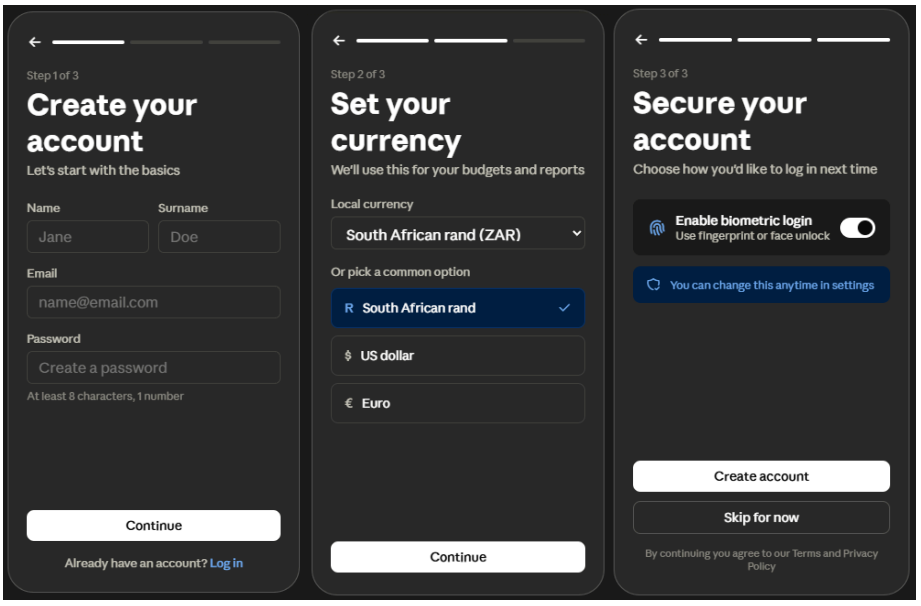
- Password/Biometric Security: Users when creating their account set up password and biometric authentication for extra security.
- A spending limit depending on budget period: Depending on the budget period (weekly, monthly, yearly) a daily, weekly, or monthly spending limit will be introduced and will show how much has been spent, how much is left to spend according to the budget period and a recommended amount the user will be allowed to spend to prevent going over the budget.
- A budget progress tracker: Progress bars will be set up for each budget plan made and a percentage will display according to how close users are to completing their budget goal
- A rewards system for meeting budget goals: Also called ‘The Fair’, when a user completes a period of time depending on the length of the budget period (daily goals for weekly budget goals, weekly goals for monthly budget goals, and monthly goals for yearly budget goals) users will receive rewards for completing the “mini budget goal” that further incentivises the user to continue their budgeting journey.
- AI integrated chat bot that can help summarise and explain the users finances, transactions, budgets and goals as well as encouraging the user to keep saving.

3. App Requirements and Mockups

Log in Screen/biometric sign in

The first screen shown to the user. If this app has been used before, they can sign in via a password, or via biometric authentication such as a thumbprint (Digital Humanity, 2026). The user is then taken to the Main Page. If no account has been created on this app, the user is redirected towards the sign up screen.



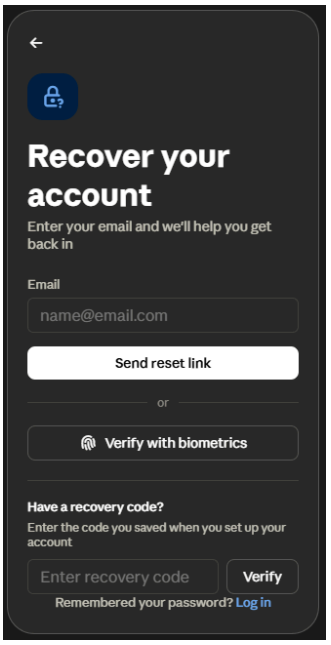


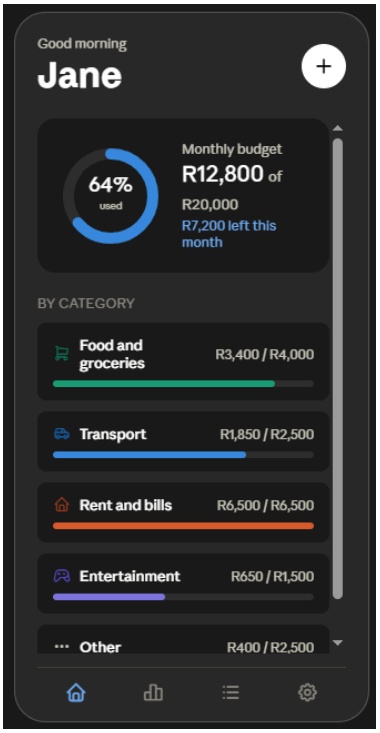
Sign Up Screen

The sign up screen has the user following the steps of creating a new account, saving a password, setting their default currency and optionally enabling biometric authentication (Monastyrska, 2024). The user is then passed on to the Main Page.

Account Recovery

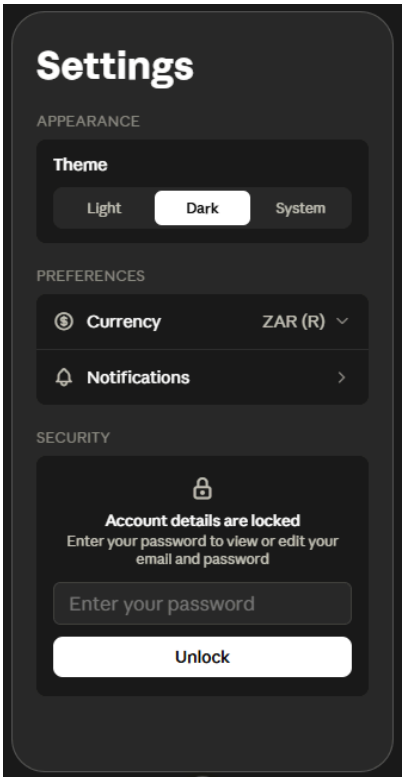
If the user forgets their passwords they can send a password reset email, the email contains a valid override password for that persons' account that is created when the user creates their account, they are assigned 5 randomly created passwords that the system will also accept as valid passwords for that email address.





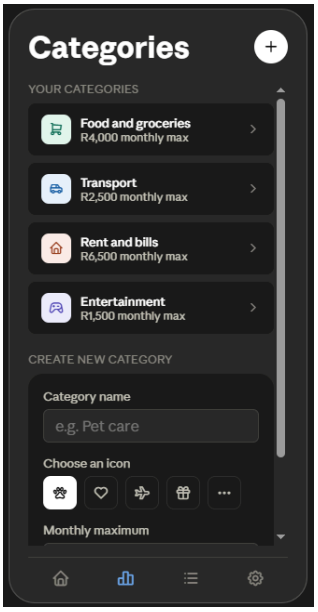
Main Page

The screen that is shown to the user immediately after logging in. Shows the user their budget, % used up and the users' daily allocation in order to stay within that budget (Paul, 2022). The main page can redirect the users to the settings and has a convenient button that allows the users to easily create new expenses (Cheredina, 2022).



Settings Page

Allows the user to set specific settings and view specific details that are locked behind a password. The user can toggle light/dark/system default mode, change the currency the system uses, after authentication, they can view the user's sensitive account details such as email and password and can change their password (Digital Humanity, 2026).



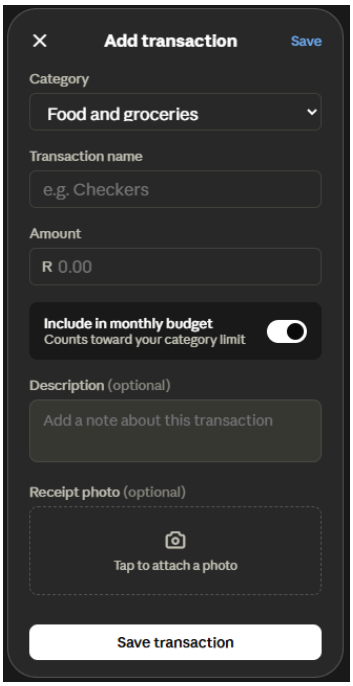
Category Creation and Viewing

Categories are used to classify expenses when the user makes one, it will be the primary factor for which the budgeting system will run on. Users will be able to create custom categories for their expenses, over and above the default ones, to remain organised (Digital Humanity, 2026). Users will be able to view all the expenses related to a specific category. i.e when "Movies" is clicked, and the app will

return all expenses from the database related to movies (Paul, 2022).

Budget Creation and Viewing

Users will be able to set maximum amounts of money that they should be allowed to spend over a selected period of time, weekly, monthly, annually (Digital Humanity, 2026). After this time frame has been completed, the budget starts again. Users will be able to view the progress through their current budget. It should show the percentage of the budget that has been spent, the amount of the budget they have available to spend, and the recommended amount of money the user will be allowed to spend each day, to the end of the budget in order to avoid going over their selected budget (Paul, 2022).

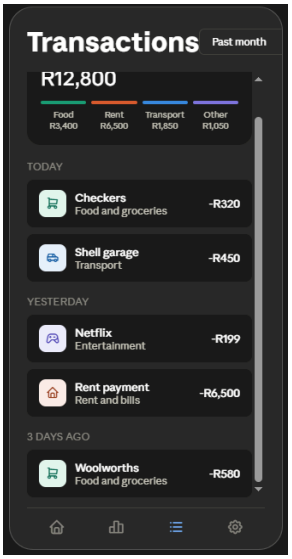


Express Expense Creation and In-depth Expense Creation

Users will have to log their expenses, they can do this in two ways, express, which streamlines the process down to only the most necessary details: The amount spent, the categories to which the expense is associated and whether or not to include this expense in the monthly budget (Digital Humanity, 2026). The user will also have the option to log expenses in a more detail oriented way (Cheredina, 2022). On top of the previous details, they can include a description of the expense, such as what specifically they bought and where they made the transaction, the date the expense occurred and an optional photo attachment, such as for a receipt or a photo of the product (Monastyrska, 2024).

Expense Viewing screen

Users can search through all their expenses to gain insights into their spending habits. They can use filters such as a date range, so that only expenses made between two dates is shown to the user, category selection, so that only expenses that are within specifically chosen categories are shown, and amount range, so that the user can only see purchases between specific amounts (Monastyrska, 2024).



Additional Features

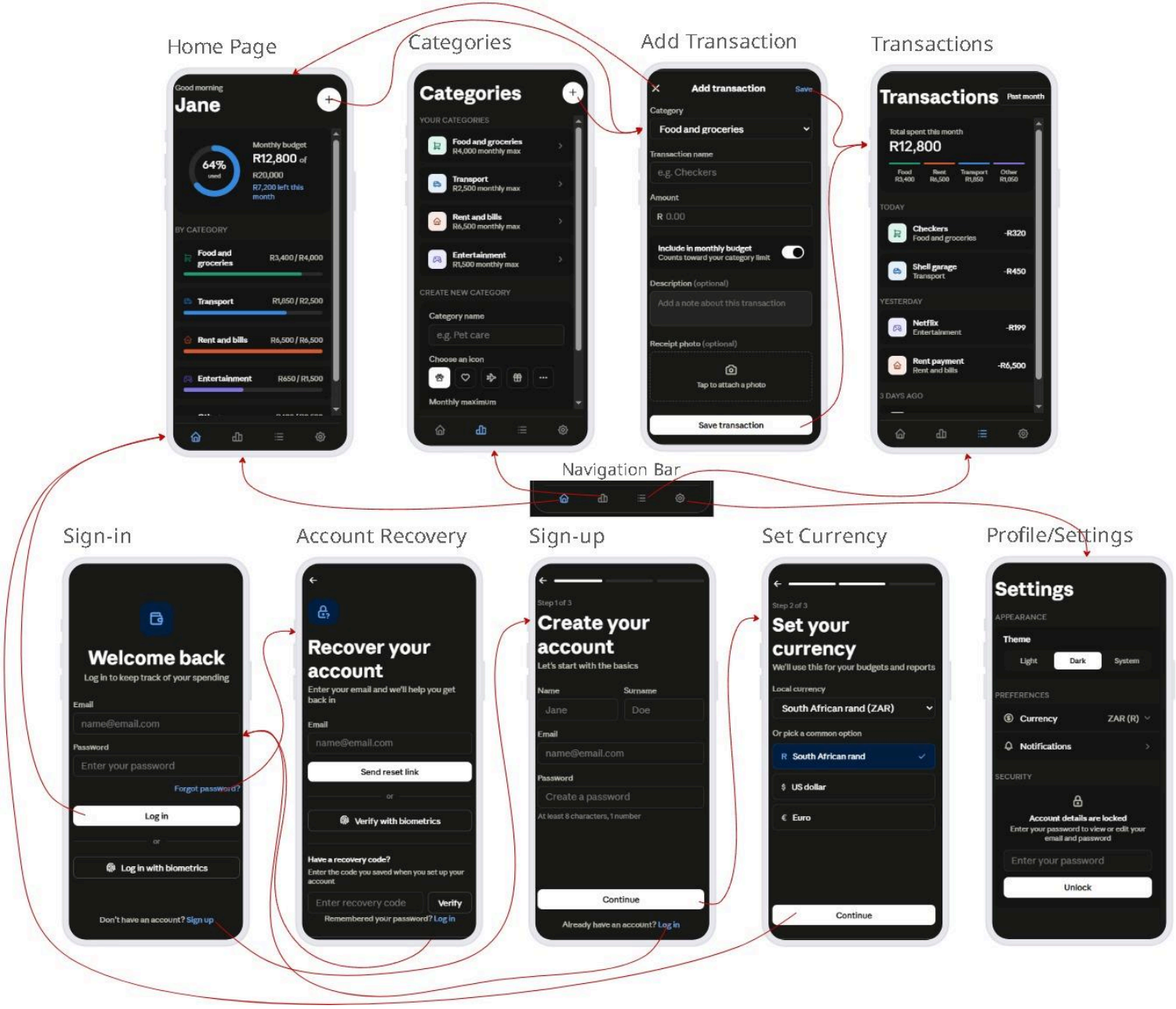
Gamification and self-rewards

If the user goes a week without exceeding their daily budget/ goes a month without exceeding their weekly budget, the player earns a ticket that they can use at the fair. The player gets to choose a game they want to play, and can win a small prize that they can buy for themself and still stay within their budget (Cheredina, 2022). I.e The user goes a month, never exceeding their weekly budget. This lets them spin a wheel/choose a tile that reveals a little reward the user can get for themself while still remaining in their budget. Eg: The user manages to say R700 under their budget by the end of the month, they choose to spin a wheel which gives the the chance to win something from R100 (most likely outcome, pizza, movie tickets, etc) to R300 (tickets to the aquarium, a nice dinner) to R500 (rarest outcome, opportunities like: rent surfing gear, go sandboarding or on a whale watching tour).

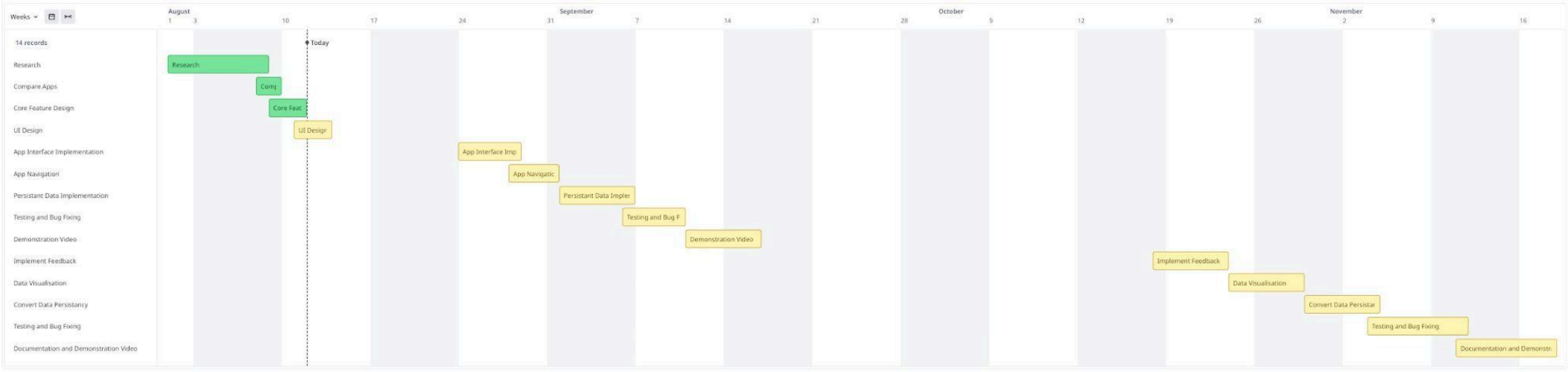
AI Insights

The app could also be the host for an embedded local AI agent, where the user could converse with a local AI model that has access to their financial records and could identify weak points in the users' spending and suggest strategies for improvement in order to guide the user towards being able to keep their spending under their chosen budget (Svitla Team, 2024).

Screen Navigation Diagram



4. Project Plan



(A separate copy of the project plan will be submitted as an addendum to improve viewability)

Conclusion

While budgeting apps are commonplace in the 21st century, those that just cater to the basic needs of their users fall short of what people need (Paul, 2022). Our budgeting app, *Republic Savings*, with a sleek and intuitive design that aims at all the needs that a user may have from a convenient budgeting app. The apps' design will aim at ease of use that emerges from informed UI and UX design, security via a biometric lock, while still allowing for expansion such as the potential for gamified user encouragement to keep them on track of their budget, and an embedded AI agent that can help users identify weak points in their spending habits and help guide them to reach their desired state of financial security.

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